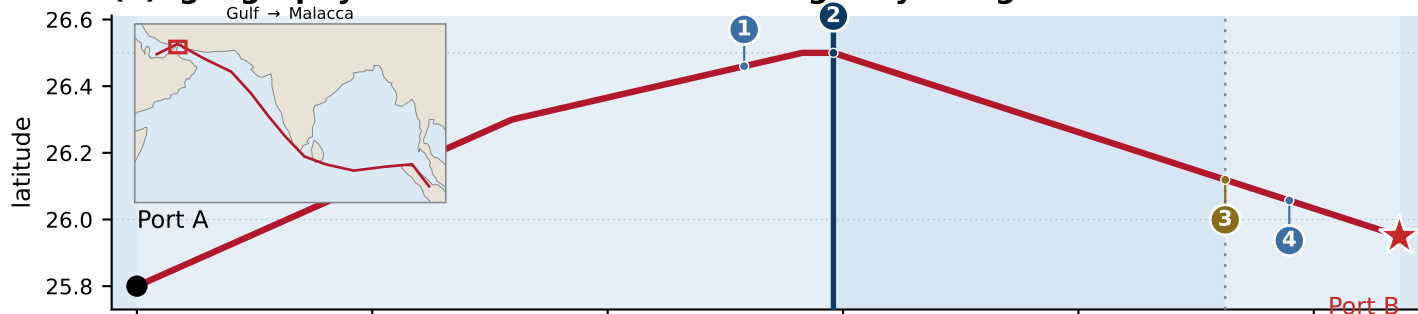


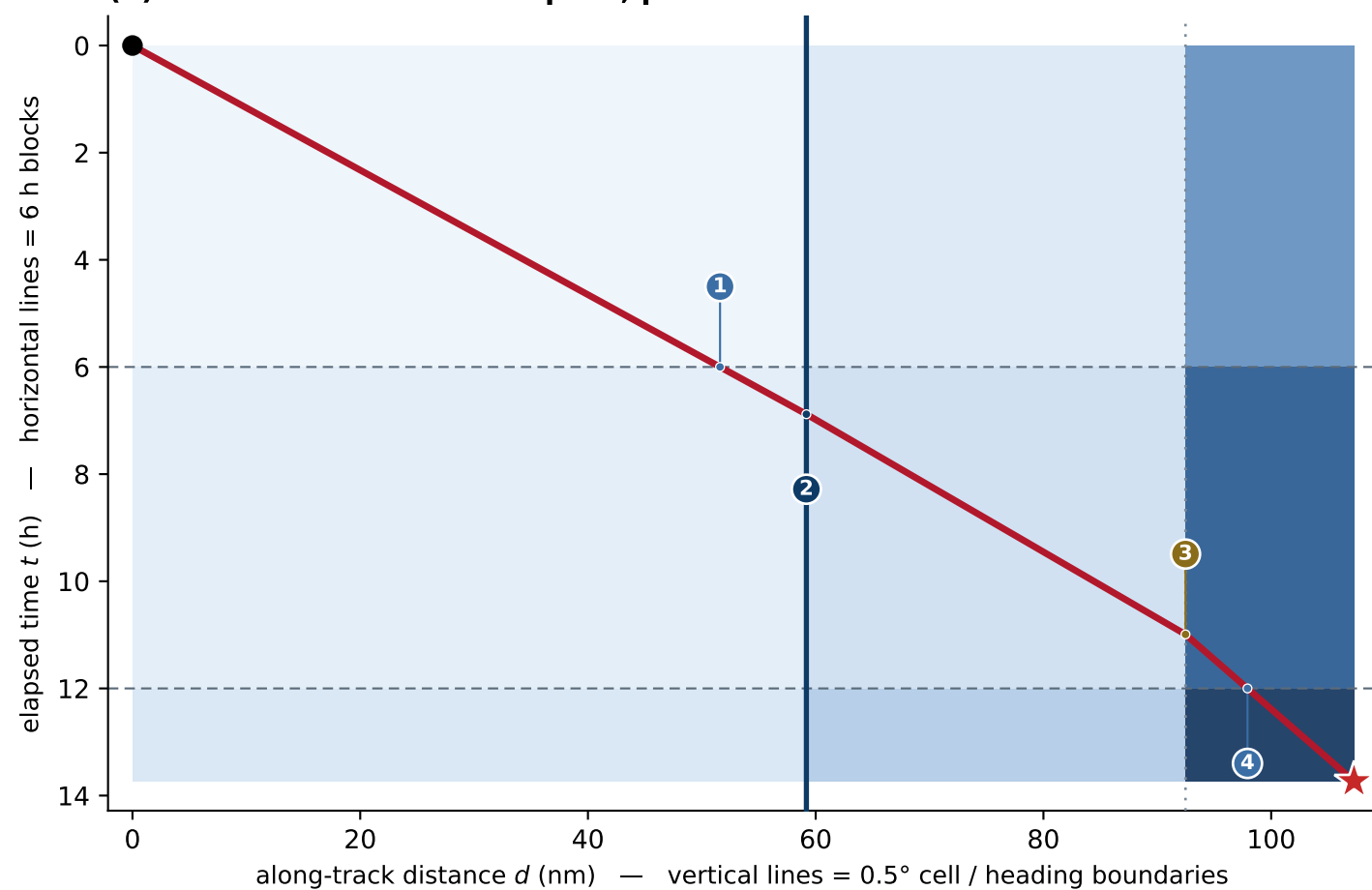
**(a) geography: the route unrolled left → right by along-track distance**



**what each point is**

- **Port A**  
departure ( $t = 0$ )
- ① **6 h block**  
weather may change (time)
- ② **heading change**  
course  $\psi$  (segment end)
- ③ **cell crossing**  
weather changes ( $0.5^\circ$ , space)
- ④ **6 h block**  
weather may change (time)
- ★ **Port B**  
arrival at the ETA

**(b) time × distance state space, painted with the sea-state field**



**reading the plane (slope = speed)**

- flat = fast (calm)
- steep = slow (rough)

